GLASGOW COLLEGE UESTC

Homework

Power Electronics (UESTC3022)

Attempt all PARTS. Total 100 marks

Power Electronics 2 Formulae List

For an inductor: $v_L(t) = L\dfrac{di_L(t)}{dt}, \quad i_L(t) = \dfrac{1}{L}\int v_L(t)dt, \quad E_L(t) = \dfrac{1}{2}L\left[i_L(t)\right]^2$

For a capacitor: $v_C(t) = \dfrac{1}{C}\int i_C(t)dt, \quad i_C(t) = C\dfrac{dv_C(t)}{dt}, \quad E_C(t) = \dfrac{1}{2}C\left[v_C(t)\right]^2$

Average value: $V_{ave} = \dfrac{1}{T}\int_0^T v(t)dt$

RMS value: $V_{rms} = \sqrt{\dfrac{1}{T}\int_0^T v(t)^2 dt}$

Duty cycle: $D \equiv \phi = \dfrac{T_{on}}{T_{on}+T_{off}} = \dfrac{T_{on}}{T_{period}}$

Amplitude modulation index of SPWM:

$$m_a = \frac{\hat{V}_{control}}{\hat{V}_{carrier}}$$

Frequency modulation index of SPWM:

$$m_f = \frac{f_{carrier}}{f_{control}}$$

Instantaneous power:

$$p(t) = \frac{v^2(t)}{R} = i^2(t)R = v(t)\times i(t) = \ force \times velocity = torque \times angular\ velocity$$

Average power:

$$P = \int_0^T v(t)i(t)dt = \frac{V_{rms}^2}{R} = I_{rms}^2 R$$

reverse recovery power loss in a diode:

$$P_{RR} = Q_{RR} \times V_{R(blocking)} \times f_{switching}$$

MOSFET switching loss:

$$P_{switching} = \frac{f_{switching} V_{DS(off)}}{2}\left(T_{on}I_{on} + T_{off}I_{off}\right)$$

Thermal circuit:

$$\Delta T = P_{\text{device}} \times \Sigma\theta_{\text{XY}} \quad \text{or} \quad \Delta T(^\circ\text{C}) = \frac{power \times time}{thermal\ capacitance}$$

Junction temperature of device:

$$T_{\text{junction}} = T_{\text{sink}} + P_{\text{device}}\left(\theta_{\text{juntion-case}} + \theta_{\text{case-sink}}\right)$$

Calculus:

$$\cos 2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$$

$$\int \cos(a\theta)d\theta = \frac{1}{a}\sin\theta \qquad \int \cos(a\theta)d\theta = -\frac{1}{a}\sin\theta$$

Q1. Answer all questions in this section:

a. A DC motor is connected to the AC mains at V_{AC} through a bridge rectifier made of silicon-controlled rectifiers (SCRs).

i. Calculate the average voltage across the motor as a function of delay angle. You may neglect the voltage drop across the SCRs. [Hint: you can relate this to a result derived in the lectures.]. [2]

ii. If the DC motor is used in the UK, what is the average voltage across the load at a firing angle of 90°? What is it if the SCRs are switched on continuously? [3]

b. Figure Q1-b shows a thyristor rectifier T which is powered by the main voltage supply u_2 and switch-controlled gate voltage supply E_g.

i. Please briefly describe the operating principle of the thyristor. [2]

ii. Please draw the voltage drop waveform (V_{load}) across the load (pure) resistor R_d, clearly indicating the delay angle and time scale. [3]

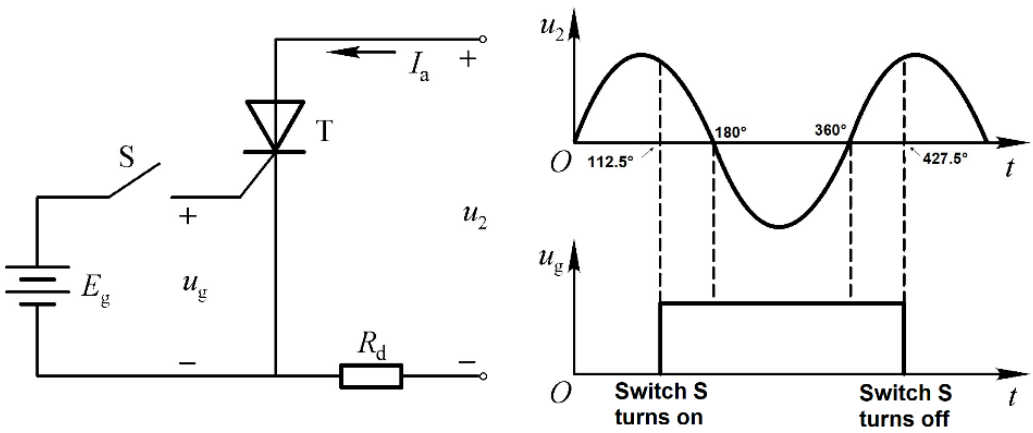


Figure Q1-b

c. Please list three advantages and three disadvantages of MOSFETs. Figure Q1-c shows the two paralleled MOSFETs T1 and T2. Please briefly describe how the positive temperature coefficient of their on-state resistance $R_{DS(on)}$ facilitates the current-sharing between the paralleled MOSFETs. [5]

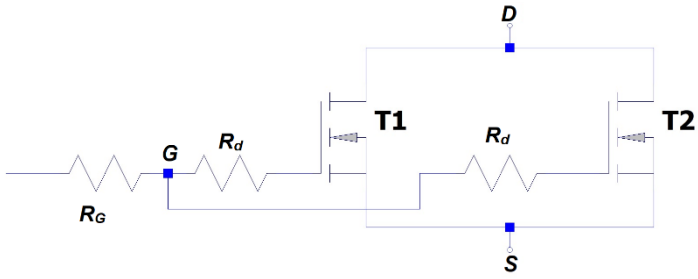


Figure Q1-c

d.

i. To avoid a risk of overheating, a TIP120 transistor is required to dissipate 40 W in an ambient temperature of 30°C. Determine the thermal resistance of the heatsink required to keep the junction temperature below 150°C. There is an insulating washer of thermal resistance 0.5°C/W between the case and the heatsink. The transistor is specified for a dissipation of 65 W below 25°C, derated at 0.5 W/°C at higher temperatures. [Note that derating is the inverse of the thermal resistance and is an alternative specification.] [3]

ii. What size of heatsink would be needed if a TIP3055 were used instead? This is a larger transistor with a maximum dissipation of 90 W derated at 0.7 W/°C. [2]

e. Consider the circuit in Figure Q1-e. Assume the power diodes are ideal, i.e., their forward voltage drops=0 V. Determine the following when a $V_{AC} = 170\sin(100\pi t)$ voltage is applied to the primary winding of the transformer:

i. Show the voltage waveforms across the secondary winding and across R_L respectively [4]

ii. What is each diode's minimum peak inverse voltage (PIV)? [1]

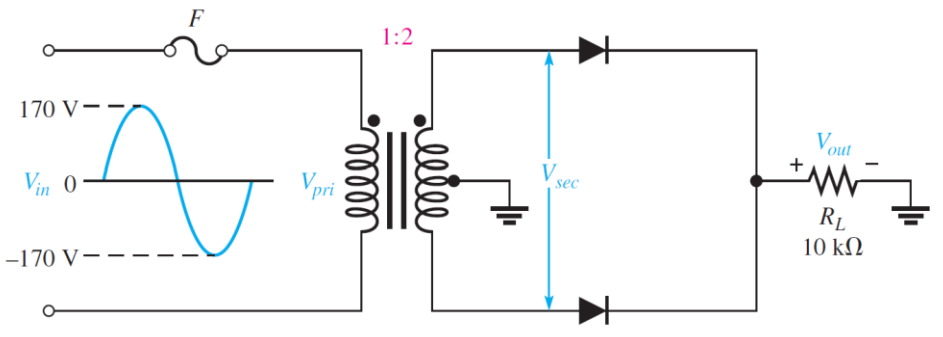


Figure Q1-e

Q2. A power supply is designed to deliver a current of 3.0 A at 15.0 V using a linear regulator. The regulator has a dropout voltage of 2.0 V and the mains input voltage is

110 V±8% at 60 Hz (North America, for example). The transformer has a regulation of 7% and is 90% efficient at full load. A bridge rectifier is used (see Fig.Q2). Assume the diode conduction angle is 30° and that the diodes drop 1.0 V at full load. The smoothing capacitor C1 is 10 mF.

a. What is the minimum AC voltage for the secondary winding of the transformer? [6]

b. What is the power dissipation in the regulator under worst-case conditions? [4]

c. What voltage rating should the capacitor have? [3]

d. What PIV (peak inverse voltage) rating should the diodes have? [2]

e. What VA rating does the transformer require? [Hint: the form factor is 2.7 for a 30° conduction angle.] [5]

f. Compared with the half-wave and the centre-tapped full-wave rectifiers, what are the advantages and disadvantages of the bridge rectifier? [5]

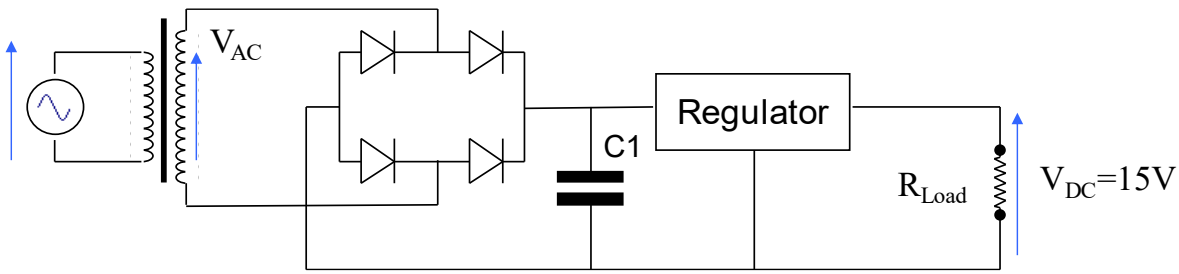


Fig.Q2

Q3. (a) The circuit diagram of a switch mode DC-DC converter is shown in Figure Q 3.a below. Consider the converter is operating in continuous mode.

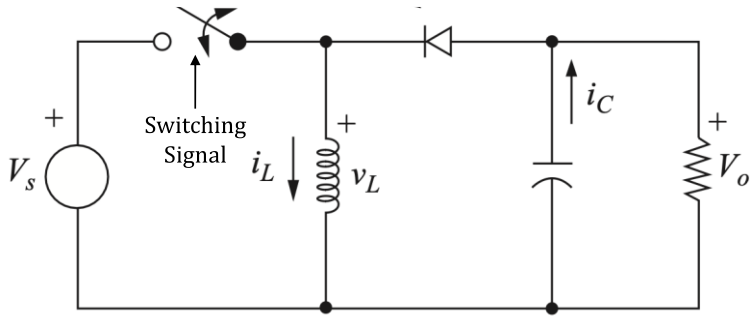


Figure Q 3.a

(i) Analyze the circuit and write down the mathematical expression for net change in current passing through the inductor over one period of switching signal in the steady state condition. Using this expression of net change in current, derive the expression for the output voltage. [9]

(ii) The DC-DC converter shown in Figure Q 3.a does not provide electrical isolation between the input and the output. Which DC-DC converter will you select to achieve this isolation? Draw the circuit diagram of the selected DC-DC converter and what will be the updated output voltage expression as compared to expression derived in part (i) of this question. [8]

(b) The circuit diagram of converter model for switching inductive load is shown below in Figure Q 3.b. The performance of the converter is to be improved using snubber circuits. Name the snubber circuits you will select for the converter and right down short reason for their inclusion. Redraw the circuit diagram below with added snubber circuits. [8]

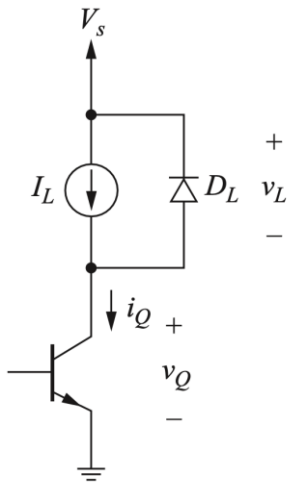


Figure Q 3.b

Q4. Figure Q4 below shows a three-phase inverter working in square-wave mode.

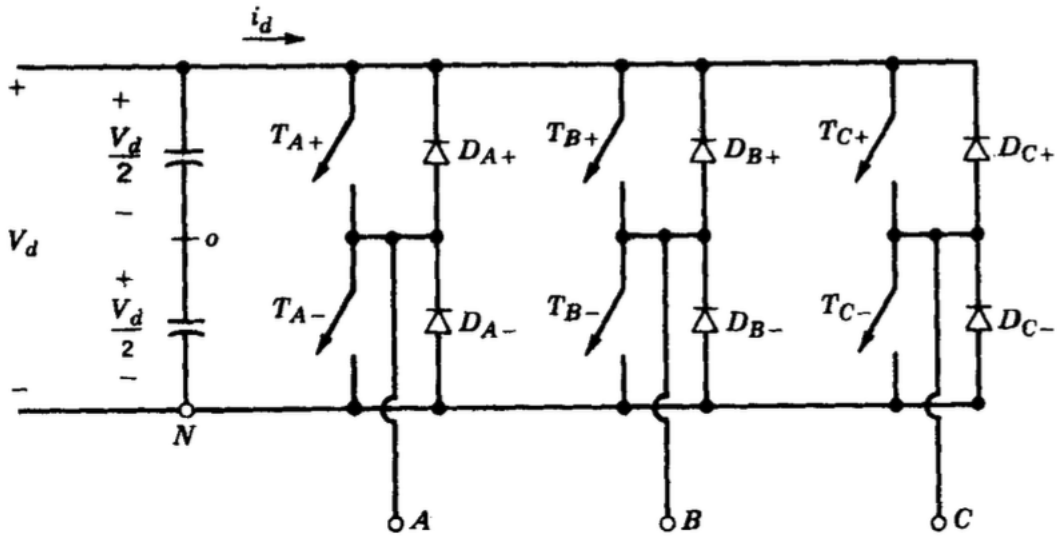


Figure Q4

a) Analyze the circuit and fill the following table based on the status of switches for each sequence number. [13]

Sequence No.	T_{A+}	T_{B+}	T_{C+}	v_{AB}	v_{BC}	v_{CA}
1	Closed	Open	Closed			
2	Closed	Open	Open			
3	Closed	Closed	Open			
4	Open	Closed	Open			
5	Open	Closed	Closed			
6	Open	Open	Closed			

b) Figure Q4.b shows a single-phase full bridge inverter.

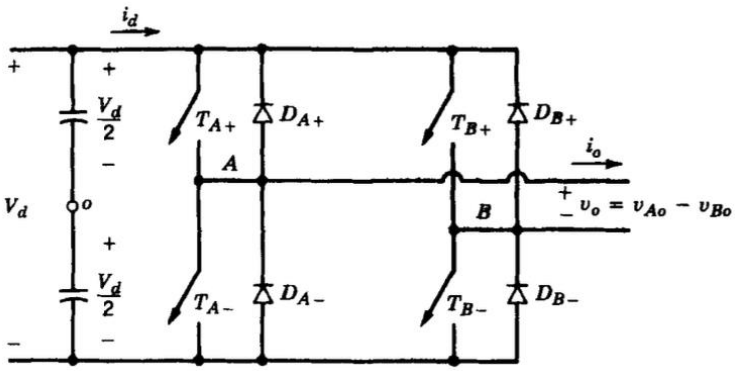


Figure Q4.b

There is a requirement that the output voltage (v_O) generated from inverter can be controlled without changing the input DC voltage (V_d) and at the same time there is a lower harmonics content in the output voltage then: [12]

i. What will be your selection of mode of operation from: **1) Square wave operation**, **2) PWM with bipolar voltage switching** and **3) PWM with unipolar voltage switching**? [2]

ii. Briefly explain the reason for your selection. [3]

iii. Draw the waveform of one or more control signals ($v_{control}$) and triangular signal (v_{tri}) (if required) used to generate the switching signal for T_{A+}, T_{A-}, T_{B+} and T_{B-}. (Note: No scaling is required while drawing waveforms). [3]

iv. Write down the conditions for T_{A+}, T_{A-}, T_{B+} and T_{B-} to be in on state for the mode of operation selected in part (i) of this question. [4]